

PRODUCT	MEMBER #	MIL	GAGE	COATING
Smart Stud	250S162-33	33	20S (Structural)	G60

250S162-33 Product Submittal

PHYSICAL PROPERTIES

Web Depth	2.5"
Flange	1.625"
Return (lip)	0.375"
Approx. Weight (lb/ft)	0.7312
Area (in^2)	0.2145
Design Thickness	0.0346
Minimum Thickness	0.0329
Yield Strength	33 KSI
Coating	G60

PRODUCT VIEW



GROSS PROPERTIES

Gross section moment of inertia about X-X axis (Ix)	0.2296
Section modulus about X-X axis (Sx)	0.1836
Radius of gyration about X-X axis (Rx)	1.034
Gross section moment of inertia about Y-Y axis (Iy)	0.0783
Radius of gyration about Y-Y axis (Ry)	0.604

EFFECTIVE / NET PROPERTIES

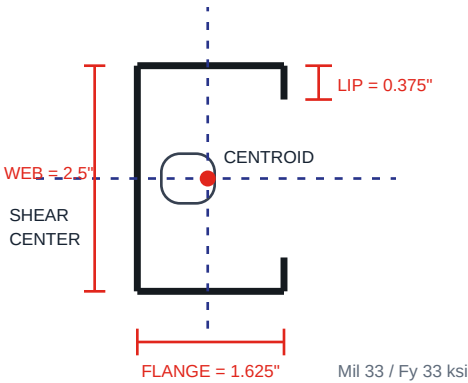
Effective moment of inertia X-axis (Ixe)	0.2120
Effective section modulus X-axis (Sxe)	0.1604
Allowable moment (Mal)	3.170
Allowable shear (Vay)	0.975

TORSIONAL PROPERTIES

St. Venant torsional constant (Jx1000)	0.0856
Torsional warping constant (Cw)	0.1131
Polar radius of gyration (Ro)	1.822

CAD SECTION

schematic dimensions



NOTES

Code Approvals & Performance Standards

- AISI S100-16 (2020) w/S2-20 North American Specification for the Design of Cold-Formed Steel Structural Members.
- AISI S240-20 North American Standard for Cold-Formed Steel Structural Framing.
 - o Compliant to ASTM C955; IBC replaced ASTM C955 with AISI S200 in IBC 2015 and AISI S240 in IBC 2018.
 - o Section A3 Material - Chemical and mechanical requirements (referencing ASTM A1003/A1003M).
 - o Section A4 Corrosion Protection (referencing ASTM A653/A653M).
 - o Section A5 Products - Thickness, shapes, tolerances, identification.
 - o Section C Installation (referencing ASTM C1007).
- AISI S202-20 Code of Standard Practice for Cold-Formed Steel Structural Framing.
 - o Section F3 Delivery, Handling, and Storage of Materials.
- IBC 2024 International Building Code.
- ICC-ES ESR-5837 Structural Studs and Track.
 - o ESR-5837 Catalog Refined Metal Technical Design Guide (10/1/2025).
- SDS for ASTM A1003 Steel Framing Products for interior framing, exterior framing, and clips/accessories.
- The technical content of this literature is effective 08/26/25 and supersedes all previous information.

DISCLAIMER

All data, details, and specifications included herein are intended as a general guide for using Refined Metal products. These products should not be used in design or construction without evaluation by a qualified engineer or architect to determine suitability for a specific use. Refined Metal assumes no liability for failure resulting from use or misapplication of computations, details, or specifications contained herein. Refined Metal assumes no liability for damages resulting from improper application or installation of these products.

DERIVED SMART STUD SPAN TABLES

Preliminary single-span estimates in feet. Verify with manufacturer span tables or sealed engineering before use.

	15 psf dead load, 40 psf live load		15 psf dead load, 60 psf live load		15 psf dead load, 125 psf live load	
Spacing OC	L/360	L/480	L/360	L/480	L/360	L/480
12	5.5	5.0	5.0	4.5	3.9	3.7
16	5.0	4.5	4.5	4.1	3.4	3.3
24	4.4	4.0	3.7	3.6	2.7	2.7

	40 psf dead load, 40 psf live load		40 psf dead load, 60 psf live load		40 psf dead load, 125 psf live load	
Spacing OC	L/360	L/480	L/360	L/480	L/360	L/480
12	4.9	4.4	4.5	4.1	3.6	3.5
16	4.4	4.0	4.0	3.7	3.1	3.1
24	3.6	3.5	3.2	3.2	2.5	2.5

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